

Unit 3

Distributions and Data-Generating Processes

Lecture notes for IB DP Biology and Chemistry

Course purpose	To help students move from raw experimental data to cautious, justified scientific claims.
Unit focus	Probability distributions, data distributions, sampling distributions, and the biological or chemical processes that make one model more plausible than another.
Central habit	Do not choose a statistical test first. First ask what kind of data were generated and what distributional model is scientifically defensible.

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1 Purpose of Unit 3

Unit 2 showed that every analysis should begin with descriptive statistics and visualisation. Unit 3 asks a deeper question:

What kind of process could have generated these data?

This is the question behind distributions. A distribution is not merely a mathematical curve. It is a compact way of describing possible values, their probabilities, and the assumptions we are making about the underlying process.

Core thesis

A distribution is a model of variation. Choosing a distribution is therefore a scientific modelling decision, not a decorative mathematical step.

By the end of this unit, students should be able to:

1. distinguish data distributions, probability distributions, and sampling distributions;
2. distinguish discrete and continuous variables;
3. connect common distributions to biological and chemical data-generating processes;
4. explain when binomial, multinomial, Poisson, normal, log-normal, and related models may be plausible;
5. explain why t, chi-square, and F distributions often describe test statistics rather than raw data;
6. treat distribution choice as a model assumption that must be justified and checked;
7. avoid the ritual phrase “the data are normal” when the actual assumption concerns residuals or sampling distributions.

2 Three meanings of distribution

The word **distribution** appears in at least three related but different ways.

Type	Question	Example
Data distribution	What pattern do my observed values show?	The actual plant heights measured by students.
Probability distribution	What values are possible under a model, and how likely are they?	If a seed has a 70% chance to germinate, the number germinating out of 20 follows a binomial model.
Sampling distribution	How would a statistic vary across repeated samples?	If many groups each sampled 20 plants and calculated a mean height, those sample means would form a distribution.

Common confusion

A student may say, “The data follow a t-distribution.” Usually that is wrong. In a t-test, the *test statistic* follows a t-distribution under the null hypothesis. The raw measurements are not

themselves t-distributed.

3 A distribution belongs to a process

A mature statistical analysis does not begin by asking, “Which test is easy in Google Sheets?” It begins by asking how the data were generated.

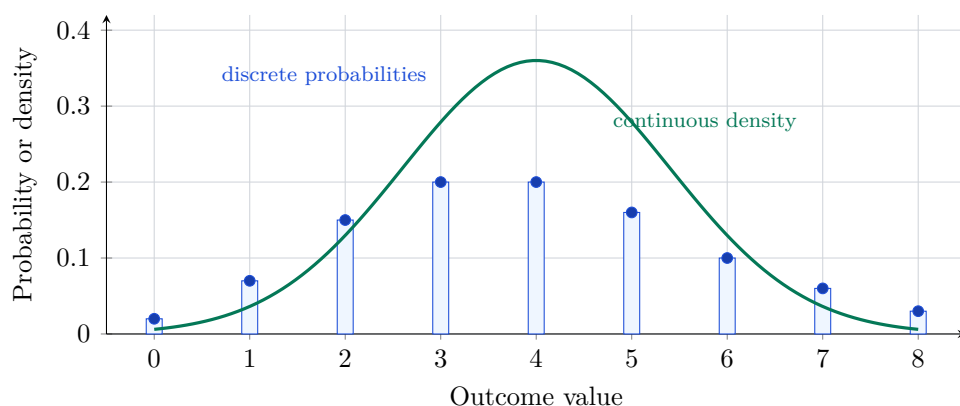


Example: bacterial colony counts may first suggest a Poisson model because colonies are count events on a plate. But if cells clump, plates differ strongly, or spreading is uneven, the counts may be more variable than the Poisson model allows. The distribution choice then tells us something scientifically useful: the process may not be independent or homogeneous.

4 Discrete and continuous data

The first branch in the decision tree is whether the outcome is **discrete** or **continuous**.

Data type	Meaning	Examples
Discrete	Countable values, often integers or categories	number of colonies, number of survivors, number of recessive offspring, blood type category
Continuous	Measured values on a scale	height, mass, pH, absorbance, concentration, reaction time, enzyme activity



Discrete distributions assign probability to individual outcomes. Continuous distributions assign probability to intervals; the probability of one exact value is usually treated as zero.

5 Bernoulli distribution: one yes/no outcome

A **Bernoulli distribution** describes one trial with two outcomes:

$$X = 1 \quad \text{success}, \quad X = 0 \quad \text{failure}.$$

If the probability of success is p , then:

$$P(X = 1) = p, \quad P(X = 0) = 1 - p.$$

Biology examples

A single seed germinates or does not germinate. A cell survives or dies. A PCR reaction is positive or negative. An offspring has a recessive phenotype or does not.

This distribution is simple, but it is the building block for many biological analyses. Every survival count, germination count, and yes/no phenotype count begins with this logic.

6 Binomial distribution: successes out of fixed trials

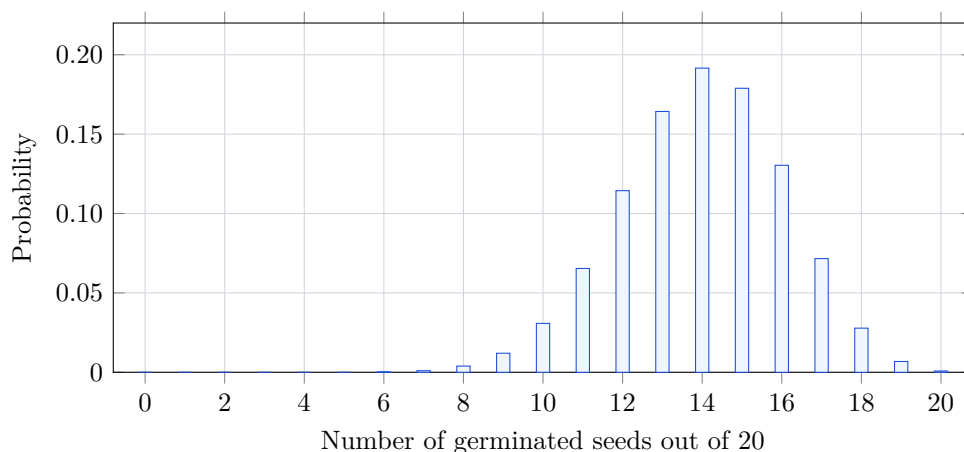
A **binomial distribution** describes the number of successes in a fixed number of independent yes/no trials.

$$X \sim \text{Binomial}(n, p)$$

where n is the number of trials and p is the probability of success in each trial.

The probability of observing exactly k successes is:

$$P(X = k) = \binom{n}{k} p^k (1 - p)^{n-k}.$$



This example assumes $n = 20$ seeds and $p = 0.70$ germination probability. The most likely result is around 14 germinated seeds, but results such as 12, 13, 15, or 16 are also plausible.

When binomial is plausible

Use binomial thinking when there is a fixed number of independent trials, each trial has two outcomes, and the probability of success is reasonably constant across trials.

When binomial may fail

If seeds come from different batches, plates have different conditions, or organisms influence each other, the independence or constant-probability assumption may fail.

7 Multinomial distribution: more than two categories

A **multinomial distribution** is the multi-category version of the binomial model. Instead of success/failure, each observation can fall into one of several categories.

Mendelian genetics

A dihybrid cross may predict four phenotype categories in a 9 : 3 : 3 : 1 ratio. Each offspring belongs to exactly one category. The vector of category counts can be modelled using a multinomial distribution under that genetic model.

This is the probability model behind many chi-square goodness-of-fit tests. The chi-square statistic is a later approximation used to judge deviation from the expected category counts.

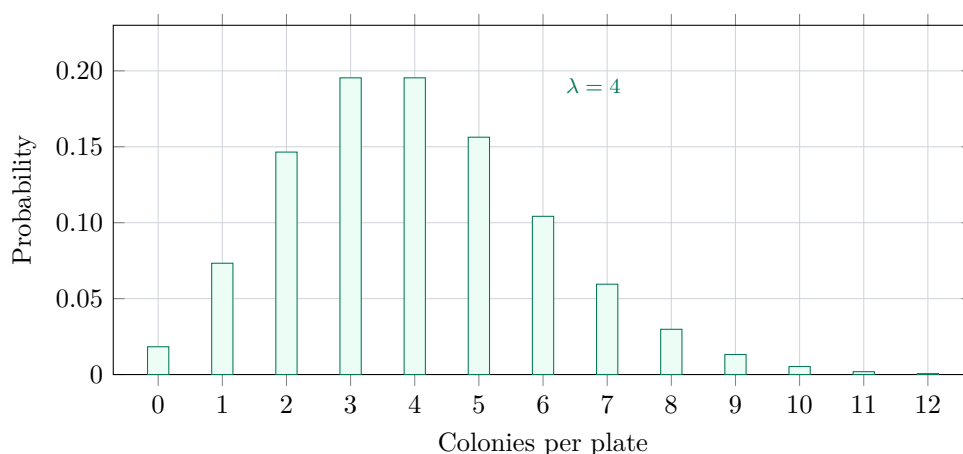
8 Poisson distribution: counts in space, time, or volume

A **Poisson distribution** is often used for counts of events in a fixed interval of space, time, area, or volume.

$$X \sim \text{Poisson}(\lambda)$$

where λ is the expected number of events in the interval. The probability of observing k events is:

$$P(X = k) = \frac{e^{-\lambda} \lambda^k}{k!}.$$



Biology examples include colonies per plate, cells per microscope field, mutations in a DNA region, insects per trap, and radioactive decay counts per minute.

A special feature of the Poisson model is:

$$\text{mean} \approx \text{variance}.$$

This gives a useful diagnostic. If colony counts have a mean near 4 but a variance near 40, a simple Poisson model is probably not adequate.

9 Overdispersion and the negative binomial model

Overdispersion means the data vary more than a proposed model expects. It is very common in biology.

Possible source	Effect on data	Example
Clumping	Some samples receive many organisms, others very few	Bacteria are not evenly suspended before plating
Environmental heterogeneity	Samples have different local conditions	Different agar plates dry at different rates
Individual variation	Organisms do not share the same probability of outcome	Some seeds are damaged before treatment
Batch effects	Groups differ for reasons unrelated to the intended variable	Different reagent bottles or incubation days

The **negative binomial distribution** is often used for count data with extra variation, especially in ecology, parasitology, microbiology, and RNA-seq analysis.

Interpretive point

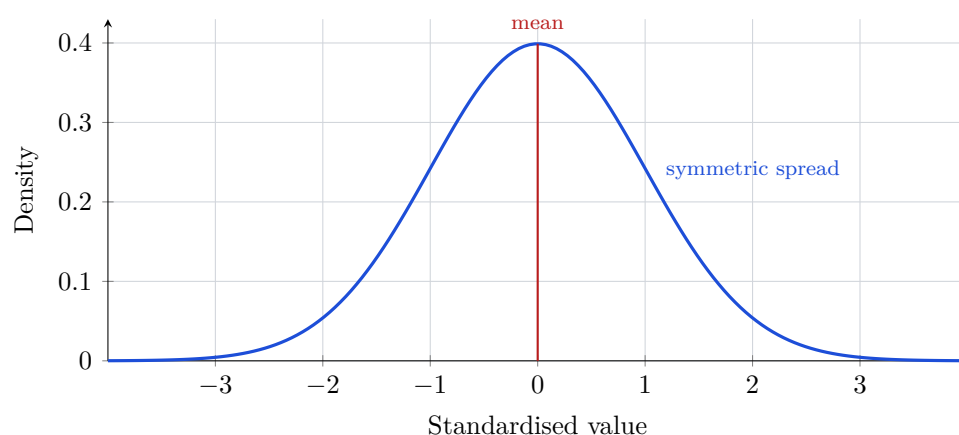
Overdispersion is not just a mathematical inconvenience. It is evidence that the biological process is less uniform or less independent than the simple model assumed.

10 Normal distribution: continuous variation around a mean

The **normal distribution** is a continuous bell-shaped distribution described by a mean μ and standard deviation σ :

$$X \sim N(\mu, \sigma^2).$$

It is plausible when variation is produced by many small, roughly independent sources of additive error or biological variation.



Where normal models are often reasonable

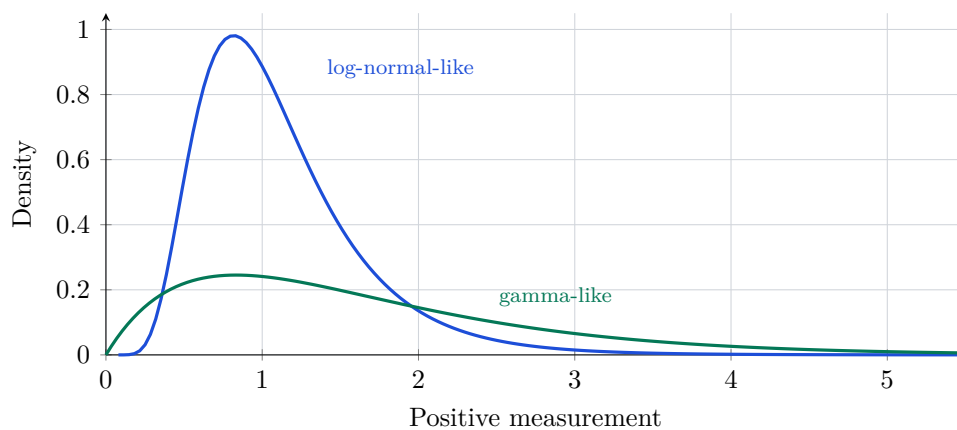
Repeated titration volumes by a stable operator, small measurement errors, plant heights within a narrow developmental stage, and residuals from a well-fitting linear model may be approximately normal.

Do not worship normality

Many biological variables are not normally distributed: counts, survival outcomes, proportions near 0 or 1, microbial abundances, hormone concentrations, and gene expression data are often skewed or discrete.

11 Log-normal and gamma distributions: positive skewed data

A variable is **log-normal** if its logarithm is approximately normally distributed. Log-normal data are positive and right-skewed: many moderate values and a few very large ones.



These distributions are useful for variables that cannot be negative and are naturally skewed.

Distribution	Useful for
Log-normal	Cell size, concentration, microbial abundance, hormone levels, some gene-expression measurements
Gamma	Waiting times, reaction times, positive rates, skewed enzyme activity values
Beta	Proportions or fractions between 0 and 1 when treated as continuous quantities

12 Sampling distributions

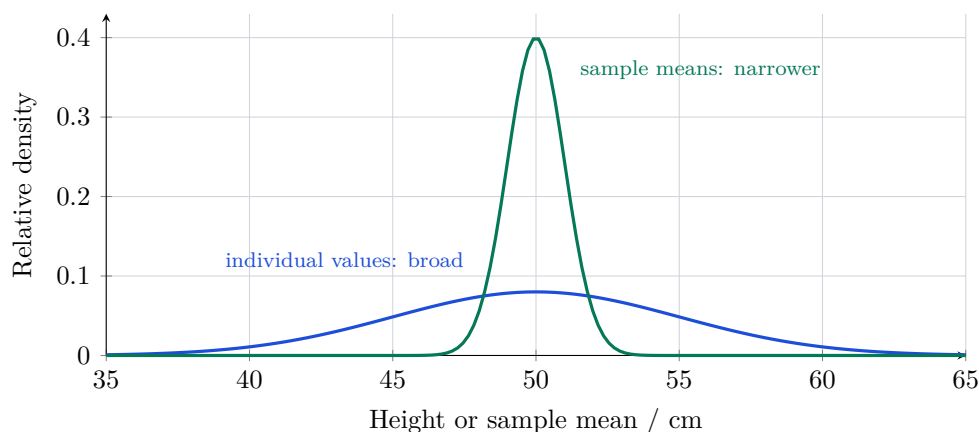
A **sampling distribution** is the distribution of a statistic across repeated samples. This is central to confidence intervals, p-values, t-tests, ANOVA, and many other methods.

Imagine the true plant-height population has standard deviation $\sigma = 5$ cm. Individual plant heights are variable. But if we repeatedly sample $n = 25$ plants and calculate each sample mean, the sample means vary less than individual plants.

$$SE_{\bar{x}} = \frac{\sigma}{\sqrt{n}}.$$

For $\sigma = 5$ and $n = 25$:

$$SE_{\bar{x}} = \frac{5}{\sqrt{25}} = 1 \text{ cm.}$$



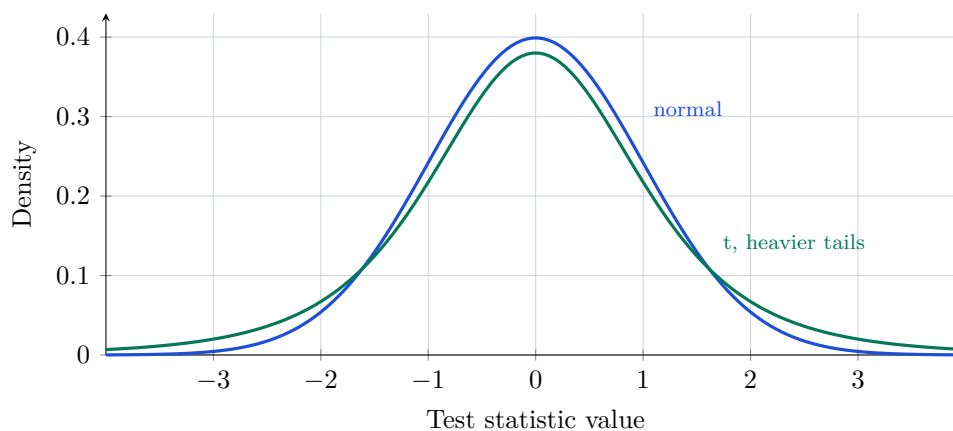
Why this matters

Inference is usually about unknown population parameters, such as a mean or proportion. The uncertainty of an estimate depends on the sampling distribution of the statistic, not only on the spread of raw observations.

13 The t-distribution: uncertainty with small samples

The **t-distribution** is used when estimating a mean from a small sample while the population standard deviation is unknown.

It looks like a normal distribution but has heavier tails. Heavy tails represent extra uncertainty.



As sample size increases, the t-distribution becomes closer to the normal distribution.

14 Chi-square and F distributions: test statistic distributions

Some distributions are mainly used as reference distributions for test statistics.

Distribution	Appears in	What it helps evaluate
Chi-square	Goodness-of-fit, tests of independence, variance inference	Whether observed counts or squared deviations are unusually large under a null model
F	ANOVA, regression, variance-ratio tests	Whether between-group variation is large compared with within-group variation

Precise language

In a chi-square goodness-of-fit test, the category counts are not “chi-square distributed.” The calculated χ^2 statistic is compared with a chi-square distribution under the null model.

15 How to choose a candidate distribution

Use the data type and the data-generating process first. Diagnostics come second.

Observation	First candidate	Question to ask
One yes/no outcome	Bernoulli	Is there one binary result?
Number of successes out of fixed trials	Binomial	Are trials independent with similar probability?
Counts in several categories	Multinomial	Are categories mutually exclusive and exhaustive?
Counts per plate, field, time, or area	Poisson	Are events independent and randomly distributed at a constant rate?
Overdispersed counts	Negative binomial	Is variance much larger than the mean?
Symmetric continuous measurement	Normal	Are residuals roughly symmetric without strong outliers?
Positive skewed measurement	Log-normal or gamma	Does the process multiply, accumulate, or produce a long right tail?
Proportion or fraction between 0 and 1	Binomial or beta	Is it based on successes out of trials, or a continuous fraction?

Mechanism before histogram

A histogram is useful, but it is not enough. A distribution should make sense for the process that generated the values.

16 Checking assumptions without ritual

Distributional assumptions can be checked, but they should not become mechanical rituals.

Useful checks include:

- raw-data plot or histogram;
- Q-Q plot for approximate normality;
- residual plot for regression or ANOVA;
- mean-variance comparison for Poisson-like count data;
- checking whether the outcome is bounded, discrete, skewed, or impossible below zero;
- sensitivity analysis: do conclusions change under another plausible model?

Shapiro-Wilk does not prove normality

A normality test with $p > 0.05$ does not prove that data are normal. It only means the test did not detect strong evidence against normality. With small samples, such tests often have low power. With huge samples, they may detect trivial departures.

17 Raw data versus residuals

Many statistical models do not require the raw response values to be normally distributed. They require the model errors or residuals to be well-behaved.

Suppose plant height depends on fertilizer dose:

$$\text{height} = a + b(\text{dose}) + \text{error}.$$

The raw heights may not be normally distributed because different doses shift the mean. What matters more is whether the residuals around the fitted relationship are approximately symmetric, independent, and not strongly patterned.

IB-level phrasing

Better than “the data were normal”: “The residual plot did not show strong curvature or unequal spread, so a linear model was considered reasonable within the tested range.”

18 Biology and chemistry examples

Research situation	Likely distributional thinking	Notes
Mendelian 3:1 phenotype counts	Binomial or multinomial	Exact binomial may be better than chi-square with small samples.
Seed germination under salt stress	Binomial	Outcome is germinated/not germinated out of fixed seeds.
Bacterial colonies per plate	Poisson or negative binomial	Check for clumping and overdispersion.
Titration volumes	Approximately normal measurement errors	Assumes technique is stable and no systematic endpoint bias.
Absorbance calibration residuals	Often approximately normal residuals	The raw absorbance values need not be normal across concentrations.
Enzyme activity across temperatures	Normal/gamma/log-normal residuals, nonlinear model	Temperature optimum means the relationship may not be linear.
Gene expression data	Log-normal or negative binomial depending on measurement type	RNA-seq read counts are commonly overdispersed counts.

19 Practice: choose the model

Task 1

For each situation, identify the likely data type and a plausible first distributional model.

1. Out of 40 seeds, 31 germinate after 48 hours.
2. A student counts bacterial colonies on 12 agar plates.
3. A colorimeter records absorbance for five replicate cuvettes at the same concentration.
4. A genetics cross produces four phenotype categories.
5. The same enzyme assay gives many small activity values and a few very large values.

Task 2

A student says: “I used a chi-square test because my data followed a chi-square distribution.” Rewrite this sentence correctly.

Task 3

A Poisson model predicts that colony counts should have mean approximately equal to variance. A class obtains a mean of 5.2 colonies per plate and a variance of 31.6. What problem does this suggest, and what might be a better modelling direction?

20 Brief answer key

1. Task 1: (1) count of successes out of fixed trials, binomial; (2) count data, Poisson first, negative binomial if overdispersed; (3) continuous measurement, normal residual/error model may be

plausible; (4) categorical counts, multinomial; (5) positive skewed continuous data, log-normal or gamma may be plausible.

2. Task 2: “I used a chi-square goodness-of-fit test because the observed category counts were compared with expected counts under a null model; the calculated chi-square statistic was then compared with a chi-square distribution.”
3. Task 3: The variance is much larger than the mean, suggesting overdispersion. A negative binomial model, or investigation of clumping/batch effects, may be more appropriate than a simple Poisson model.

21 Unit 3 summary

What students should retain

A distribution is a model of variation produced by a process. The correct question is not “Which test do I remember?” but “What kind of data were generated, what assumptions are plausible, and what conclusions remain valid if those assumptions are checked?”

The essential decisions are:

1. Identify the data type: categorical, count, proportion, or continuous measurement.
2. Think about the generating process: independent trials, events in space, measurement error, multiplicative growth, clustering, or skew.
3. Choose a candidate distribution that matches both the data type and the process.
4. Check whether the data or residuals strongly contradict the model.
5. Report conclusions conditionally and cautiously.